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I confirm that I understand my coursework needs to be submitted online via mySecondTeacher Platform under the relevant module page before the deadline in order for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1. Create a directory structure as shown below and show it in the terminal.

To create the directory as given in the question I used p option. The option basically means parent and it helps to make complex directory structure in just one line which helps to save time. In the given structure there is a parent directory Dev which has four different sub-directories and they are: projects, scripts, backups, logs. The syntax is: "mkdir -p Dev/{projects,scripts,backups,logs}".

```
bhumika@B:~$ mkdir -p Dev/{projects,scripts,backups,logs}
bhumika@B:~$ tree Dev
Dev
├── backups
├── logs
├── projects
└── scripts

5 directories, 0 files
bhumika@B:~$
```

Figure 1: Creating a directory

2. Inside projects/, create a two subdirectories alpha and beta using a relative path.

To make two subdirectories inside projects, we have to use mkdir command.

For alpha, the syntax is "mkdir Dev/projects/alpha".

For beta, the syntax is "mkdir Dev/projects/beta".

```
bhumika@B:~$ mkdir Dev/projects/alpha
bhumika@B:~$ mkdir Dev/projects/beta
bhumika@B:~$ tree Dev
Dev
├── backups
├── logs
├── projects
│   ├── alpha
│   └── beta
└── scripts

7 directories, 0 files
bhumika@B:~$
```

Figure 2: Creating two subdirectories inside projects

3. Create an empty file readme.md in alpha/ using an absolute path.

To create an empty file, we use touch command. The syntax for creating an empty file with an absolute path is "touch /home/bhumika/Dev/projects/alpha/readme.md".

```
bhumika@B:~$ touch /home/bhumika/Dev/projects/alpha/readme.md
bhumika@B:~$ tree Dev
Dev
├── backups
├── logs
├── projects
│   ├── alpha
│   │   └── readme.md
│   └── beta
└── scripts

7 directories, 1 file
bhumika@B:~$
```

Figure 3: Creating a file named readme.md

4. Use the touch command with an absolute path to create a file named deploy.sh in the script directory.

To create a file named deploy.sh using the touch command with an absolute path in the script directory, we have to write the syntax "touch /home/bhumika/Dev/scripts/deploy.sh".

```
bhumika@B:~$ touch /home/bhumika/Dev/scripts/deploy.sh
bhumika@B:~$ tree Dev
Dev
├── backups
├── logs
├── projects
│   ├── alpha
│   │   └── readme.md
│   └── beta
└── scripts
    └── deploy.sh

7 directories, 2 files
bhumika@B:~$
```

Figure 4: Creating a file named deploy.sh

5. Display the detailed permission and ownership of the deploy.sh file.

To display the detailed permission and ownership of the deploy.sh file, we have to use ls command with l option. The l option is used to list the file in long format and it contains permission and ownership details as well. The syntax is "ls -l Dev/scripts/deploy.sh".

```
bhumika@B:~$ ls -l Dev/scripts/deploy.sh
-rw-r--r-- 1 bhumika bhumika 0 Jan 20 07:24 Dev/scripts/deploy.sh
bhumika@B:~$
```

Figure 5: displaying the permission and ownership of deploy.sh

6. Set the permission on deploy.sh so that the owner has rwx permissions and others have not

To give the rwx permission to only owner and not for rest, we have to use chmod command. For setting such permission on deploy.sh file, we have to write command "chmod 700 Dev/scripts/deploy.sh".

```
bhumika@B:~$ chmod 700 Dev/scripts/deploy.sh
bhumika@B:~$ ls -l Dev/scripts/deploy.sh
-rwx------ 1 bhumika bhumika 0 Jan 20 07:24 Dev/scripts/deploy.sh
bhumika@B:~$
```

Figure 6: Setting the permission

7. Set permissions of beta to:

- Users: RWX
- Group: RX
- Others: None

To set the permission given in the question above for beta, we have to use syntax "chmod 750 Dev/projects/beta".

```
bhumika@B:~$ chmod 750 Dev/projects/beta
bhumika@B:~$ ls -l Dev/projects
total 8
drwxr-xr-x 2 bhumika bhumika 4096 Jan 20 07:23 alpha
drwxr-x--- 2 bhumika bhumika 4096 Jan 20 07:21 beta
bhumika@B:~$
```

Figure 7: Setting the permission for beta

8. Copy alpha from projects/ to logs/

Since the alpha contains the file readme.md, we have to use r command which will recursively copy both file to logs. The syntax to do so is "cp -r Dev/projects/alpha Dev/logs/alpha".

```
bhumika@B:~$ cp -r Dev/projects/alpha Dev/logs/alpha
bhumika@B:~$ tree Dev
Dev
├── backups
├── logs
│   └── alpha
│       └── readme.md
├── projects
│   ├── alpha
│   │   └── readme.md
│   └── beta
└── scripts
    └── deploy.sh

8 directories, 3 files
bhumika@B:~$
```

Figure 8: Copying alpha

9. Rename the copied file in logs/ to alpha_old.sh.

For renaming the file, I have utilized the move command mv. The syntax for renaming the file copied in logs to alpha_old.sh is "mv Dev/logs/alpha Dev/logs/alpha_old.sh".

```
bhumika@B:~$ mv Dev/logs/alpha Dev/logs/alpha_old.sh
bhumika@B:~$ tree Dev
Dev
├── backups
├── logs
│   └── alpha_old.sh
│       └── readme.md
├── projects
│   ├── alpha
│   │   └── readme.md
│   └── beta
└── scripts
    └── deploy.sh

8 directories, 3 files
bhumika@B:~$
```

Figure 9: Renaming the file

10. Delete the beta/ directory from projects/.

To delete the beta directory from projects, we have to write the syntax "rmdir Dev/projects/beta". The command rmdir is used to delete empty directory.

```
bhumika@B:~$ rmdir Dev/projects/beta
bhumika@B:~$ tree Dev
Dev
├── backups
├── logs
│   └── alpha_old.sh
│       └── readme.md
├── projects
│   └── alpha
│       └── readme.md
└── scripts
    └── deploy.sh

7 directories, 3 files
bhumika@B:~$
```

Figure 10: Deleting the beta directory